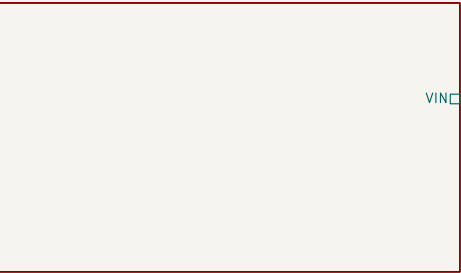


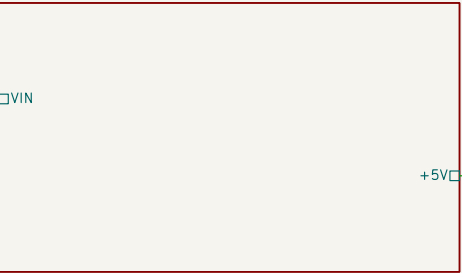
CAMERA CONTROLLER – signal flow

GND is a global net on every sheet. Double-click a block to open it.

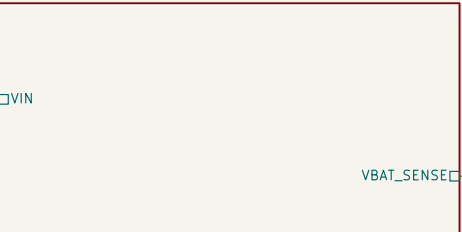
1. Battery 2S + protection



2. Buck to 5 V



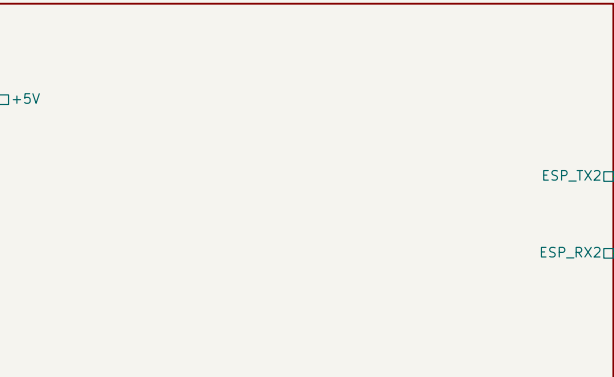
3. Pack voltage sense



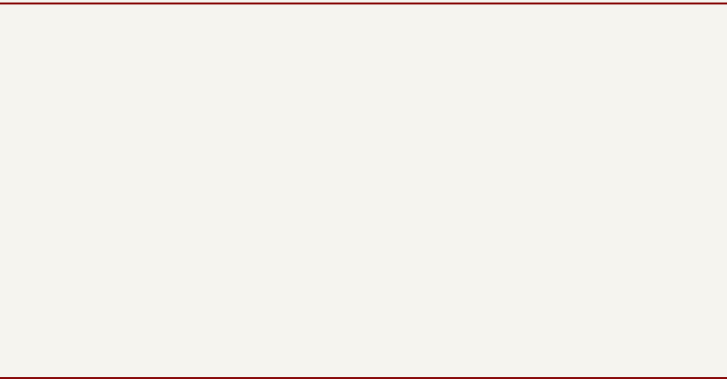
4. ESP32 socket



5. RunCam Split 4

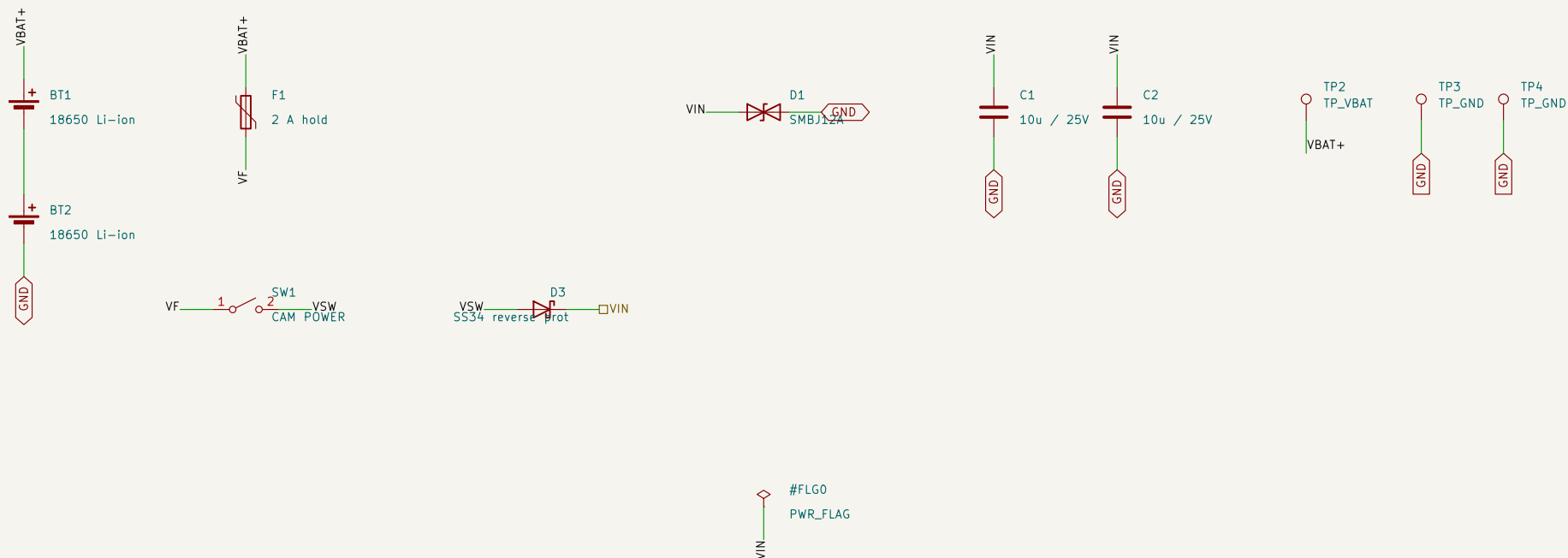


6. COTS / SRAD supplies



Block 6 is electrically isolated from the rest of this board on purpose.

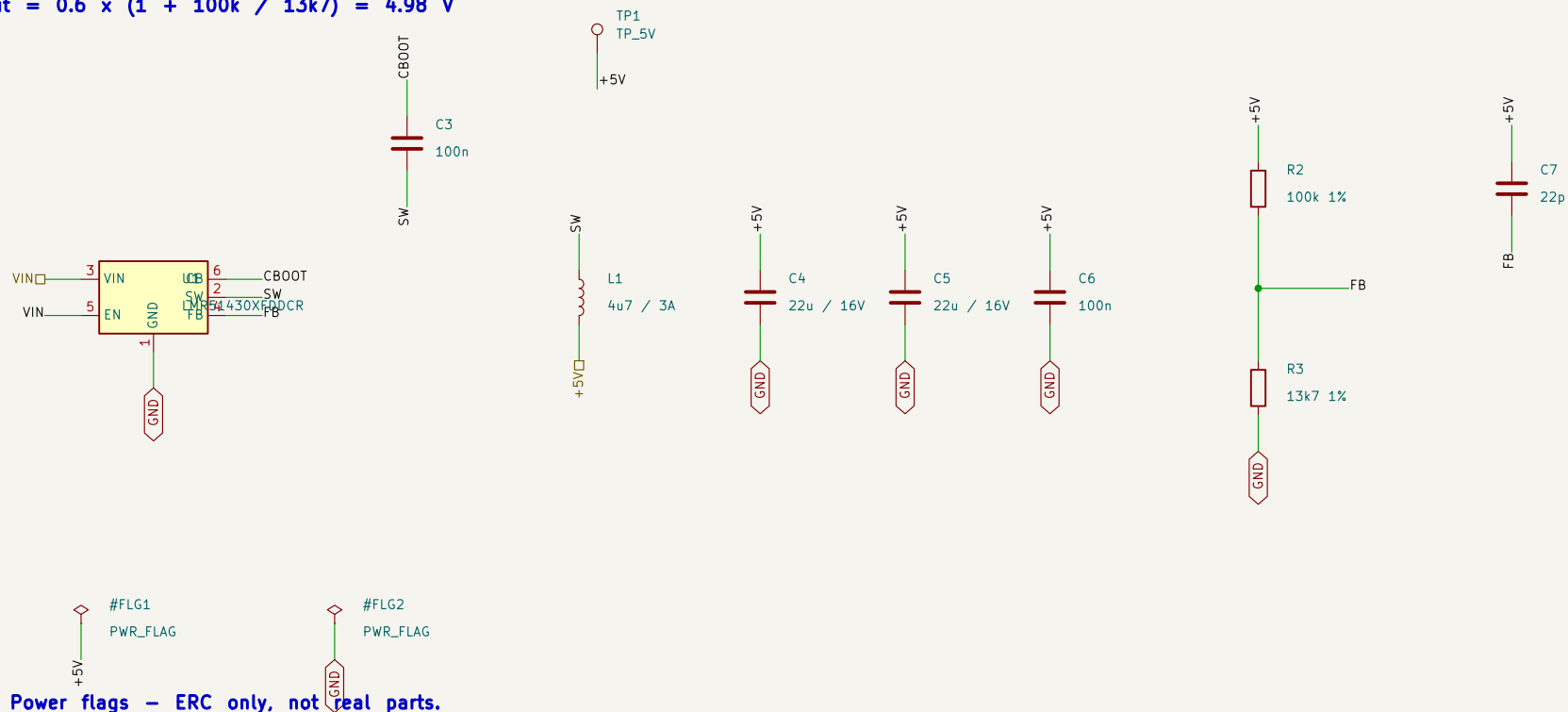
2 x 18650 in series -> 7.4 V nom, 8.4 V max



Power flag – ERC only.

Stanford SSI		
Sheet: /1. Battery 2S + protection/		
File: power_in.kicad_sch		
Title: 1. Camera + ESP32 supply (2S)		
Size: A4	Date:	Rev: A
KiCad E.D.A. 10.0.0		Id: 2/7

LMR51430 $V_{out} = 0.6 \times (1 + 100k / 13k7) = 4.98 \text{ V}$



Power flags – ERC only, not real parts.

Stanford SSI

Sheet: /2. Buck to 5 V/

File: buck_5v.kicad_sch

Title: 2. Buck 8.4 V $\rightarrow$ 5.0 V

Size: A4	Date:
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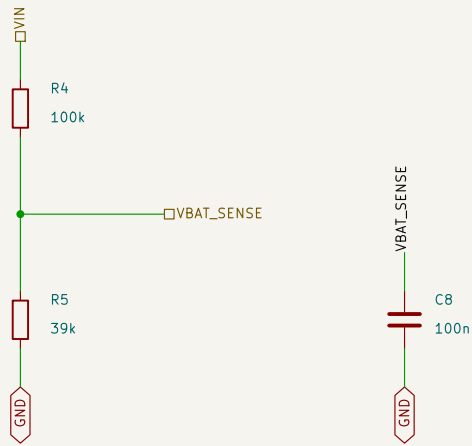
SIZE: A1	D
KiCad E.D.A. 10.0.0	

Date:

Rev: A

Id: 3/7

8.4 V x 47 / 147 = 2.69 V at the ADC



Stanford SSI

Sheet: /3. Pack voltage sense/
File: sense.kicad_sch

Title: 3. Pack voltage sense

Size: A4

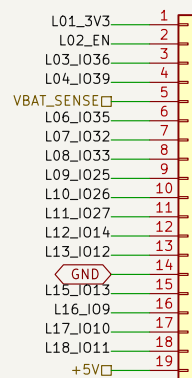
Date:

Rev: A

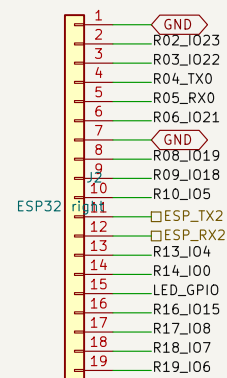
KiCad E.D.A. 10.0.0

Id: 4/7

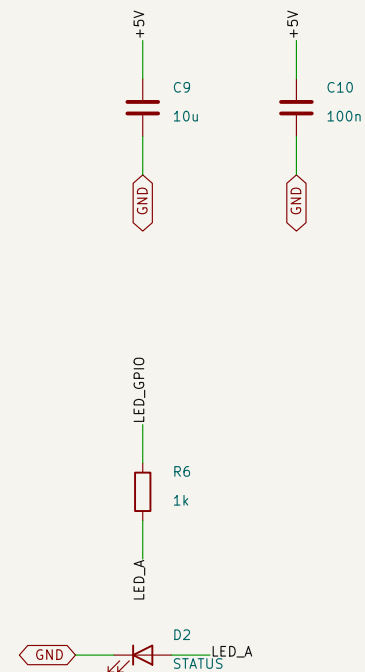
2 x 1x19 female header. GPIO6–11 go to the module SPI flash – do not use.



J1
ESP32 left



ESP32 right



Stanford SSI

Sheet: /4. ESP32 socket/

File: esp32.kicad_sch

Title: 4. ESP32-DevKitC V4 socket

Size: A4

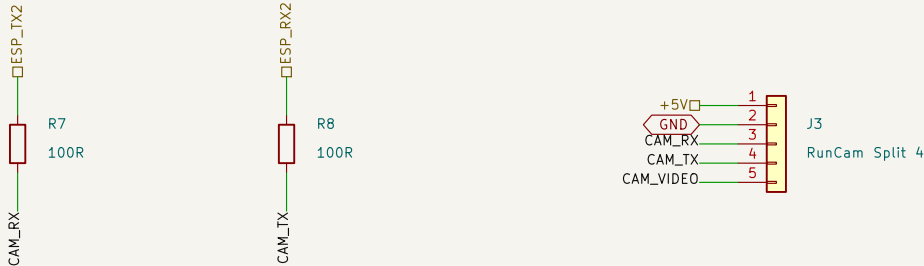
Date:

Rev: A

KiCad E.D.A. 10.0.0

Id: 5/7

5–20 V in. Single 5–pin harness: power, ground and UART.



Stanford SSI		
Sheet: /5. RunCam Split 4/		
File: camera.kicad_sch		
Title: 5. RunCam Split 4		
Size: A4	Date:	Rev: A
KiCad E.D.A. 10.0.0		Id: 6/7

Own cell, own switch, per DTEG 6.9.1. SW2 / SW3 take the pin-switch leads; they sit next to the pin-switch stack. All connectors latch (Micro-Fit 3.0).

These returns are deliberately NOT tied to GND – that independence is the point.

